Datatype

September 8, 2026

1 TUPLE

Tuple is immutable

[1]:
```
Tuple = (0,1,2,3,4,5)
```

[2]:
```
Tuple[3]
```

[2]: 3

[3]:
```
Tuple[:3]
```

[3]: (0, 1, 2)

[4]:
```
Tuple[1:6:2]
```

[4]: (1, 3, 5)

[5]:
```
Tuple[0] = 10
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[5], line 1
----> 1 Tuple[0] = 10

TypeError: 'tuple' object does not support item assignment
```

[6]:
```
Tuple[5] + 5
```

[6]: 10

[7]:
```
5 in Tuple
```

[7]: True

[8]:
```
1 not in Tuple
```

[8]: False

[9]:
```
Tuple2 = (5,6,7)
```

[16]:
```
Tuple = (0,1,2,3,4,5)
Tuple2 = (5,6,7)
print(Tuple + Tuple2)
```

```
(0, 1, 2, 3, 4, 5, 5, 6, 7)
```

[ ]:
```
tup1ele = (10,) #single element tuple should end with comma
```

[ ]:
```
test =(1,[1,2])
```

[ ]:
```
test[1] = [1,2,3]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[12], line 1
----> 1 test[1] = [1,2,3]

TypeError: 'tuple' object does not support item assignment
```

[ ]:
```
test[1][1] = [1,2,3]
```

[ ]:
```
test
```

[ ]:
```
(1, [1, [1, 2, 3]])
```

2 LIST

List is immutable

[37]:
```
numl = [1,2,3,4,5,6]
strl = ['a','b','c']
mixl = [1,2,6,45,'a','b'] # Mix datatype list
```

[ ]:
```
numl[2] is 10
```

```
<>:1: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
<>:1: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
C:\Users\AV\AppData\Local\Temp\ipykernel_19120\1534824144.py:1: SyntaxWarning:
"is" with 'int' literal. Did you mean "=="?
  numl[2] is 10
```

[ ]:
```
False
```

[ ]:
```
strl[0] in mixl
```

[ ]: True

[ ]:
```
str1[0] = 'b'
str1[1:] = ['c','d']
```

[ ]:
```
str1
```

[ ]: ['b', 'c', 'd']

[ ]:
```
str1.insert(0,'a')
```

[ ]:
```
str1
```

[ ]: ['a', 'b', 'c', 'd']

[ ]:
```
str1.append('e')
```

[ ]:
```
str1
```

[ ]: ['a', 'b', 'c', 'd', 'e']

[54]:
```
str12 = ['f','g','h','i','i']
```

[ ]:
```
str1.extend(str12)
```

[ ]:
```
str1
```

[ ]: ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'i']

[ ]:
```
str1.remove('i')
```

[ ]:
```
str1
```

[ ]: ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i']

[ ]:
```
str1.pop(2)
```

[ ]: 'c'

[50]:
```
str1
```

[50]: ['a', 'b', 'd', 'e', 'f', 'g', 'h', 'i']

[ ]:
```
del str12[2]
print (str12)
```

['f', 'g', 'i', 'i']

[ ]:
```
str12.clear()
```

[ ]:
```
strl2
```

[ ]: []

[ ]:
```
del strl2
```

[ ]:
```
strl2
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[60], line 1
----> 1 strl2

NameError: name 'strl2' is not defined
```

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

2.1 Challenge

Even no from list

[ ]:
```
nol = [1,2,3,4,5,6,7,8,9,10]
```

[29]:
```python
for i in nol:
    if i % 2 == 0:
        print(i, end=" ")
```

```
2 4 6 8 10
```

[ ]:
```python
# List Comprehension
even = [x for x in nol if x % 2 == 0]
print (even)
```

```
[2, 4, 6, 8, 10]
```

Listt Comprehention

`[ 'var' for 'var' in 'LIST' 'Condition' ]`

2.2 Challenge

Leap Year from list

[27]:
```python
year = [2000,1900,2024,2004,2026]
```

[28]:
```python
for i in year:
    if i % 100 ==0:
        if i % 400 == 0:
            print(i, "is a leap year")
        else:
            print(i, "is a not leap year")
    else:
        if i % 4 == 0:
                    print(i, "is a leap year")
        else:
            print(i, "is a not leap year")
```

```
2000 is a leap year
1900 is a not leap year
2024 is a leap year
2004 is a leap year
2026 is a not leap year
```

[ ]:
```python

```